Local link replacement with explicit boundary conditions

Research note and proof audit

October 9, 2026

Abstract

We examine the rough-region replacement step suggested by Family 215's spin-model argument. An unconditional extension that preserves arbitrary frozen gauge boundary links while removing every rough plaquette is false: we exhibit an open set of frustrated one-link boundaries. We prove instead an excess-energy tail estimate for every boundary, and an explicit measurable gauge-covariant replacement kernel for a sufficient class of compatible boundaries. For n edited links in four-dimensional $SU(2)$ Wilson theory at inverse coupling $H \geq 1$, the latter kernel has density at most $(24H^{3/2})^n$, makes every affected plaquette curvature at most $4/\sqrt{H}$, and gives a conditional energy–entropy bound $\exp[-kp^2/2 + n(48 + \log 24 + \frac{3}{2}\log H)]$ for k plaquettes of curvature greater than $p/\sqrt{H}$. The compatibility condition is explicit, gauge invariant, sufficient but not necessary. No estimate is supplied for the probability of incompatible boundaries, for arbitrary renormalized interactions, or for cutoff comparison. This is not a continuum construction or a mass-gap proof, and no novelty claim is made.

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1 Finite lattice and the exact task

Work on a four-dimensional rectangular periodic lattice whose periods are at least four. Each positively oriented link e has a variable $U_e \in \mathrm{SU}(2)$; its reverse has variable U_e^{-1}. Each unoriented

elementary plaquette is counted once with a fixed orientation. Set

$$s(g) = 1 - \tfrac{1}{2}\operatorname{Tr} g, \qquad E(U) = \sum_p s(U_p), \qquad \mu_H(\,\mathrm{d}U) = Z_H^{-1} e^{-HE(U)} \prod_e \mathrm{d}U_e, \quad H \geq 1, \tag{1}$$

with probability Haar measure on every link. The trace of an $SU(2)$ matrix is real. The nonnegative-action convention here differs from the unshifted Wilson convention only by an extensive constant.

Fix, *before integrating*, a nonempty set J of editable links, and write $n = |J|$, $x = U_J$, $z = U_{J^c}$. Let $P(J)$ consist of all plaquettes that touch at least one link in J, and let $q_p = |\partial p \cap J|$. Define

$$E_J(x, z) = \sum_{p \in P(J)} s(U_p).$$

Each link belongs to six plaquettes, so

$$|P(J)| \leq 6n, \qquad \sum_{p \in P(J)} q_p = 6n, \qquad \sum_{p \in P(J)} q_p^2 \leq 24n. \tag{2}$$

For an open finite complex with all affected plaquettes included, the corresponding statements are inequalities. Only E_J changes when x is replaced. The conditional law given z is proportional to $e^{-HE_J(x,z)}\,\mathrm{d}x$.

The original spin replacement in [1] combines exterior compatibility, a small-density-cost sampling kernel, and a lower bound on original energy proportional to the edited volume. These are distinct requirements. In a gauge theory one must edit *links*, not independent plaquette matrices; this automatically respects all algebraic identities among plaquettes, but does not guarantee small plaquette curvature.

2 Geometry and Haar-volume constants

Use the chordal operator metric $d(g, h) = \|g - h\|_{\mathrm{op}}$. It is bi-invariant and obeys

$$d(\mathbf{1}, g_1 \cdots g_m) \leq \sum_{j=1}^{m} d(\mathbf{1}, g_j), \qquad d(g^{-1}, h^{-1}) = d(g, h).$$

If the eigenvalues of g are $e^{\pm i\theta}$, $0 \leq \theta \leq \pi$, then

$$d(\mathbf{1}, g) = 2\sin(\theta/2), \qquad s(g) = \tfrac{1}{2} d(\mathbf{1}, g)^2. \tag{3}$$

Let C_ρ be the geodesic cap $\theta \leq \rho$, for $0 < \rho \leq 1$. Every $g \in C_\rho$ obeys $d(\mathbf{1}, g) \leq \rho$ and is e^X for a traceless anti-Hermitian X with $\|X\|_{\mathrm{op}} \leq \rho$. Probability Haar measure is uniform measure on the unit three-sphere, so

$$b(\rho) := \mathrm{Haar}(C_\rho) = \frac{2}{\pi} \int_0^\rho \sin^2\theta \,\mathrm{d}\theta = \frac{\rho - \sin\rho\cos\rho}{\pi} \geq \frac{\rho^3}{24}. \tag{4}$$

The lower bound follows from $\sin\theta \geq \theta/2$ for $0 \leq \theta \leq 1$ and $\pi < 4$. These deliberately loose constants suffice.

3 Why arbitrary frozen boundaries cannot be filled smoothly

Proposition 3.1 (A robust one-link obstruction)**.** *There is a frozen exterior for one editable link in the lattice* (1) *such that $E_J(U, z) = 6$ for every $U \in \mathrm{SU}(2)$. For every U, at least three affected plaquettes obey $d(\mathbf{1}, U_p) \geq \sqrt{2}$. Thus no replacement preserving that exterior can make all affected plaquettes have curvature less than any $t < \sqrt{2}$. The impossibility persists on an open set of frozen exterior data.*

Proof. For one oriented link from x to y, the six adjacent plaquettes can, using trace cyclicity and inversion, be written with energies $1 - \frac{1}{2}\operatorname{Tr}(US_j^{-1})$, where S_j is transport from x to y along the other three edges of that plaquette. These six staple paths are internally edge-disjoint when the periods are at least four. Set three staple transports to $+\mathbf{1}$ and three to $-\mathbf{1}$, assigning central signs on a distinct edge of each path and identity on the rest. These are actual link assignments, so plaquette constraints hold exactly. Other, unaffected plaquettes may be rough; their contribution is constant in the conditional law of the edited link.

Write $a = \operatorname{Tr} U/2 \in [-1, 1]$. The six energies are three copies each of $1 - a$ and $1 + a$, with sum six. If $a \geq 0$, the latter three are at least one; if $a \leq 0$, the former three are. Apply (3). The conditional law of U is exactly Haar, independently of H.

If each staple remains within chordal distance δ of its assigned sign, bi-invariance and the reverse triangle inequality show that for every U the same three large-curvature plaquettes have curvature at least $\sqrt{2} - \delta$. For fixed $t < \sqrt{2}$ choose $0 < \delta < \sqrt{2} - t$. Staple transports depend continuously on frozen links, so this condition contains an open neighborhood of the displayed boundary. Every nonempty open cylinder of link data has positive Haar measure, and positive probability at every finite H under the finite Wilson law. No lower bound on that probability uniform in H is asserted. □

Proposition 3.2 (An explicit one-link boundary-defect quantity)**.** *Identify $SU(2)$ with unit quaternions. For the six frozen staple transporters let $s_j \in S^3 \subset \mathbb{R}^4$, let $B = \sum_{j=1}^{6} s_j$, and put*

$$\mathcal{D}(z) = \sum_{i<j} d(S_i, S_j)^2.$$

Then the boundary-forced minimum is exactly

$$e_*(z) = 6 - |B| = \frac{\mathcal{D}(z)}{6 + |B|}, \qquad \frac{\mathcal{D}(z)}{12} \leq e_*(z) \leq \frac{\mathcal{D}(z)}{6}. \tag{5}$$

The quantity $\mathcal{D}$ is gauge invariant and consists of frozen-loop holonomy mismatches, since $d(S_i, S_j) = d(\mathbf{1}, S_i S_j^{-1})$.

Proof. The fundamental normalized trace of US_j^{-1} is the Euclidean inner product $u \cdot s_j$, so $E_J(U, z) = 6 - u \cdot B$ and minimization over $|u| = 1$ gives $6 - |B|$. Chordal operator distance of two quaternion matrices is their Euclidean quaternion distance. Expanding the pairwise squares gives $\mathcal{D} = 36 - |B|^2$, proving the identity and the bounds using $0 \leq |B| \leq 6$. Gauge transformations act on all staples by the same left and right unit quaternions, hence orthogonally on $\mathbb{R}^4$. □

This identifies a boundary quantity to retain or charge in the simplest case. It does not show that it is statistically rare or decompose the multi-link minimum into independent one-link quantities. Minimizing aggregate energy also need not minimize the largest individual curvature.

The obstruction is boundary frustration, not an appeal to continuum instanton topology. An unrestricted finite lattice has a connected product of $SU(2)$ link spaces. A gauge transformation

given at finitely many boundary vertices can always be assigned arbitrary values at the remaining vertices. A nonzero-degree continuous map $S^3 \to SU(2)$ therefore does *not*, by itself, prohibit a discrete link filling. Continuum extension obstructions become relevant only with additional regularity, interpolation, admissibility, or sector constraints, which must be specified separately.

4 An arbitrary-boundary excess-energy estimate

Set $e_*(z) = \min_x E_J(x, z)$; the minimum exists by compactness. It is continuous in z, since the uniform difference of the two energy functions bounds the difference of their minima.

Lemma 4.1 (A cap near a conditional minimizer)**.** *For every frozen z, choose any minimizer V of $E_J(\cdot, z)$. For $x_e = V_e e^{X_e}$ with $\|X_e\|_{\mathrm{op}} \le \rho$,*

$$E_J(x, z) \le e_*(z) + 12 \sum_{e \in J} \|X_e\|_{\mathrm{op}}^2 \le e_*(z) + 12n\rho^2.$$

Proof. Take the simultaneous path $x_e(t) = V_e e^{tX_e}$. A reversed occurrence has inverse $e^{-tX_e} V_e^{-1}$. Differentiating a plaquette product twice and using unitary operator norms gives

$$\left| \frac{\mathrm{d}^2}{\mathrm{d}t^2} s(U_p(t)) \right| \le \left(\sum_{e \in \partial p \cap J} \|X_e\|_{\mathrm{op}} \right)^2 .$$

Here $|\operatorname{Tr} A|/2 \le \|A\|_{\mathrm{op}}$, including nonsymmetric complex matrices. Cauchy–Schwarz and the incidence bound imply

$$|E_J''(t)| \le \sum_p q_p \sum_{e \in \partial p \cap J} \|X_e\|^2 \le 24 \sum_{e \in J} \|X_e\|^2 .$$

The first derivative at zero is zero because V minimizes on the smooth compact product manifold. Integrate the second derivative with weight $1 - t$ from zero to one. □

Theorem 4.2 (Uniform excess above the boundary minimum)**.** *For every frozen exterior z and $h \ge 0$,*

$$\mu_H\{E_J \ge e_*(z) + h \mid z\} \le \min\{1, \exp[-Hh + n(12 + \log 24 + \tfrac{3}{2} \log H)]\}. \tag{6}$$

Proof. At $\rho = H^{-1/2}$, the product cap in Lemma 4.1 has Haar mass $b(\rho)^n$ and energy at most $e_*(z) + 12n/H$. It gives the conditional partition-function lower bound $b(\rho)^n e^{-He_*(z) - 12n}$. The event numerator is at most $e^{-H(e_*(z)+h)}$, since its Haar mass is at most one. Divide and apply (4). This pointwise argument uses no measurable choice of a minimizer as z varies. □

This theorem does not bound $e_*(z)$ itself. In Proposition 3.1 it equals six and every link value is already a minimizer. The theorem is therefore compatible with rough plaquettes having conditional probability one.

5 A sufficient boundary compatibility condition

Let F_J be the set of nonedited links occurring on plaquettes in $P(J)$, and let W_J contain all endpoints of links on those plaquettes. Form the frozen graph (W_J, F_J), including isolated vertices. Choose a spanning forest, one root per component, and path orientations by a deterministic ordering depending

only on J and lattice labels, never on link values. For vertex x let $T_x(z)$ be ordered transport from its component root to x along the tree, and set $h_x = T_x^{-1}$. For each frozen link $e = (x, y)$ define

$$R_e(z) = h_x^{-1} z_e h_y, \qquad \eta_J(z) = \max_{e \in F_J} d(\mathbf{1}, R_e(z)), \tag{7}$$

with the maximum over an empty set defined as zero. Tree residuals are identity. All other residuals are fundamental-cycle holonomies.

Lemma 5.1 (Meaning and covariance of compatibility)**.** *The function η_J is continuous and gauge invariant. The condition $\eta_J = 0$ says exactly that every frozen-graph cycle has identity holonomy, equivalently that there are vertex frames satisfying $z_{xy} = h_x h_y^{-1}$ on all frozen links. For every $\eta > 0$, the event $\eta_J(z) < \eta$ contains a nonempty open neighborhood of identity frozen data.*

Proof. For site gauges g_x, path transports transform as $T_x(z^g) = g_{r(x)} T_x(z) g_x^{-1}$. Thus $h_x(z^g) = g_x h_x(z) g_{r(x)}^{-1}$ and every frozen residual transforms by conjugation at its component root. Continuity and metric invariance prove the first claim. Tree paths and fundamental cycles prove the flat claim directly. At identity data all residuals are identity; continuity proves the last assertion. □

The condition depends on the chosen forest quantitatively, not through gauge choice. It is sufficient, not necessary, for low-curvature filling. A long loop can have nontrivial holonomy while each individual plaquette in a spanning surface is small. Exact flat data may be a measure-zero class; the positive-tolerance version below avoids relying on that class. No estimate on how often a useful tolerance is satisfied is supplied.

6 A gauge-covariant link replacement kernel

Theorem 6.1 (Compatible-boundary cap replacement)**.** *For $0 < \rho \le 1$ and fixed J, there is a Borel replacement kernel $K_z(x, \mathrm{d}y)$ with these properties:*

1. *All links outside J stay exactly fixed. The conditional law of the new edited links y_J depends only on the frozen links in F_J, not on the original x. The full-output law additionally copies the exterior.*
2. *Its conditional density in y_J with respect to product Haar is at most $b(\rho)^{-n} \le (24\rho^{-3})^n$.*
3. *Under every site gauge transformation it pushes forward to the kernel for the transformed data.*
4. *For every output and each $p \in P(J)$,*

$$d(\mathbf{1}, y_p) \le q_p \rho + (4 - q_p)\eta_J(z). \tag{8}$$

If $\eta_J(z) \le \rho$, then every affected curvature is at most 4ρ and $E_J(y, z) \le 48n\rho^2$. If $\eta_J(z) = 0$, the sharper bound $E_J(y, z) \le 12n\rho^2$ holds.

Proof. For each connected component C of the frozen graph choose an independent Haar frame A_C. Put $S_x = h_x A_{C(x)}$. Independently choose each Q_e, $e \in J$, with normalized Haar restricted to C_ρ, and set

$$y_e = S_x Q_e S_y^{-1} \quad (e = (x, y) \in J), \qquad y_e = z_e \quad (e \notin J). \tag{9}$$

All maps are finite products of continuous maps and Borel cap indicators, so the law is a Borel kernel. Given the component frames, left/right Haar invariance gives joint density at most $b(\rho)^{-n}$; averaging frames preserves that bound. There is no division by a gauge-orbit volume.

Under a gauge transform couple $A_C^g = g_{r(C)} A_C$ and keep the same Q_e. The transformed frames satisfy $S_x^g = g_x S_x$, while the coupled root frames retain their independent Haar laws. Thus $y_{xy}^g = g_x y_{xy} g_y^{-1}$, proving covariance.

In the S frames, edited links equal Q_e and each frozen link equals $A_C^{-1} R_e A_C$. The metric triangle inequality around a plaquette gives (8), including reversed edge orientations. If $\eta_J \le \rho$, use $s(y_p) \le (4\rho)^2/2$ and $|P(J)| \le 6n$. For exact flat data, use $s(y_p) \le q_p^2 \rho^2/2$ and (2). □

Since the outputs are link configurations, all plaquette identities hold automatically. Every loop consisting entirely of unedited links is preserved exactly. No claim is made about preservation of loops traversing J, an instanton number defined on an admissible interpolation, or any sector label not included in the frozen data. No reflection-positivity claim is made for the measure pushed forward by this kernel.

7 The integrated energy–entropy estimate

The following elementary measure argument is the useful replacement for an unjustified invertible "surgery." It tracks lost information by a density bound, not by asserting a change-of-variables inverse.

Lemma 7.1 (Bounded-density replacement)**.** *Let $x = (x_J, z)$ have product probability reference measure. Let $\mathcal{Q} \ge 0$ be measurable with $Z_{\mathcal{Q}} = \int e^{-HE(x)} \mathcal{Q}(x)\,\mathrm{d}x \in (0, \infty)$. Suppose a probability kernel changes only J, has conditional density at most M, and for all original x in a measurable event $\mathcal{A}$ and almost all outputs y obeys $E(x) - E(y) \ge h$ for one fixed real h. Then*

$$\frac{1}{Z_{\mathcal{Q}}} \int_{\mathcal{A}} e^{-HE(x)} \mathbb{E}_{K_x}[\mathcal{Q}(y)]\,\mathrm{d}x \le M e^{-Hh}. \tag{10}$$

For $\mathcal{Q} = 1$ the left side is $\mu_H(\mathcal{A})$. For the probability under the $\mathcal{Q}$-weighted law, additionally assuming $\mathcal{Q}(x) \le e^{Ln} \mathcal{Q}(y)$ on these pairs yields $\mu_{H,\mathcal{Q}}(\mathcal{A}) \le e^{Ln} M e^{-Hh}$.

Proof. On the kernel support replace $e^{-HE(x)}$ by $e^{-Hh} e^{-HE(y)}$. Bound the kernel density by M and then discard all restrictions on the original x_J and the frozen z. Tonelli's theorem applies to the nonnegative integrand. Integrating the original x_J contributes at most one; the remaining integral in (y_J, z) is $Z_{\mathcal{Q}}$. For the last statement use the extra comparison of $\mathcal{Q}(x)$ and $\mathcal{Q}(y)$ before applying the same argument. □

No probability claim for an arbitrary $\mathcal{Q}$ follows by silently identifying $\mathcal{Q}(x)$ with $\mathcal{Q}(y)$. Signed or complex perturbations require an appropriate positive base weight and a separate absolute-value/control argument. This lemma does not supply the renormalized model's interaction bounds or its covering expansion.

Let $p \ge 4$, let k be a positive integer, and set $\rho = H^{-1/2}$, $t = p/\sqrt{H}$. Define the event, for fixed J,

$$\mathcal{A}_{J,k} = \{\eta_J(z) \le H^{-1/2}, \quad |\{p' \in P(J) : d(\mathbf{1}, U_{p'}) > t\}| \ge k\}.$$

When $t \ge 2$ the rough-plaquette event is empty; the interesting regime is $p < 2\sqrt{H}$.

Theorem 7.2 (Compatible rough-plaquette suppression)**.** *For each boundary with $\eta_J(z) \le H^{-1/2}$, the conditional probability of at least k rough affected plaquettes has the following bound. The same bound holds for the joint event $\mathcal{A}_{J,k}$ under the finite Wilson law:*

$$\mu_H(\mathcal{A}_{J,k}) \le \min\{1, \exp[-kp^2/2 + nc_H]\}, \qquad c_H = 48 + \log 24 + \tfrac{3}{2} \log H. \tag{11}$$

The exponential bound, without truncation by one, holds for the replacement integral (10) with any nonnegative $\mathcal{Q}$ and this event. That integral need not be a probability. All originally rough affected plaquettes are removed by the output, and no affected plaquette exceeds t. Unaffected plaquettes do not change. For exactly flat compatible boundary data, replace 48 by 12 in the conditional bound.

Proof. On the event, (3) gives $HE_J(x,z) \geq kp^2/2$. Theorem 6.1 gives $HE_J(y,z) \leq 48n$ and density at most $(24H^{3/2})^n$. Unaffected energies cancel exactly. Apply Lemma 7.1. The output curvature is at most $4/\sqrt{H} \leq t$. For a fixed boundary the same proof uses only product Haar on J. □

The signed/nonlocal factors generated by renormalization are not assumed to meet this theorem's raw-Wilson energy bound. The displayed general $\mathcal{Q}$ integral is a restricted interface, not a proof that they do.

8 Edited volume, counting, and what is still missing

One bad plaquette does not imply original energy proportional to the number of edited links. For example, in identity background set one link to $-\mathbf{1}$ and take J to be an arbitrarily large connected link set containing it. Only six plaquettes are nonidentity, and $E_J = 12$ independently of $|J|$. Thus the density cost cannot be charged to edited volume without a separate geometric energy-density statement.

Corollary 8.1 (A scoped animal bound)**.** *Suppose an event for each fixed connected J imposes its compatibility condition and at least $|J|/s$ rough affected plaquettes, where $s \geq 1$. If $p^2 \geq 4sc_H$, its probability is at most $\exp[-|J|p^2/(4s)]$. Connect two lattice links when they share a plaquette. For a fixed root link and $\kappa = p^2/(4s) > 2\log 18$, the probability of at least one such connected event containing that link is at most*

$$\frac{e^{-\kappa}}{1 - 18^2 e^{-\kappa}}.$$

Proof. Apply (11) with $k \geq n/s$ and absorb nc_H using the assumed budget. The link adjacency graph has degree at most $6 \cdot 3 = 18$. Every connected set of n links containing the root has a deterministically chosen spanning-tree traversal of length $2(n-1)$; the traversal determines the visited set. Thus there are at most $18^{2(n-1)}$ such sets. The union bound and a geometric series prove the claim; no independence of the region events is used. □

For $p = (\log H)^P$ and $s = (\log H)^q$, the elementary absorption is available for large H if $2P > q + 1$. This numerical budget is meaningful only after the asserted compatible regions and bad-plaquette density have actually been obtained.

Disjoint edited link sets do not suffice for additive local estimates when their affected plaquettes overlap. If the sets $P(J)$ are disjoint, energies and product-kernel density costs can be combined; otherwise one must use the union with its actual affected plaquettes. For regions chosen from the original configuration, freeze their geometry in each integration term and sum possible geometries explicitly. The animal bound covers the listed events, not an unconstructed decomposition of all rough configurations.

Unresolved inputs for the cutoff-comparison program

1. A local region-selection or defect-label scheme must handle typical *incompatible* boundaries. The frustrated example cannot simply be excluded; its probability or alternative retained energy must be bounded.

2. Such a scheme must supply enough original energy per edited link and controlled counts of regions and boundary labels. The raw one-link estimate and the forest condition do not do this automatically.

3. Exact renormalization generates interactions beyond the original Wilson sum. Their changes under replacement, locality, and signed covering weights need independent uniform estimates.

4. Only after those estimates could one seek a cutoff-comparison error small on a fixed physical box. Continuum observables and their reconstruction remain additional tasks, as in the first research note.

The present results settle the proposed *unrestricted local filling* in the negative and give two precise restricted replacements. They do not establish the nonperturbative cutoff-comparison lemma or resolve Yang–Mills existence and mass gap.

9 Verification scope

The accompanying audits independently check the boundary obstruction, frame covariance and density, and integrated estimates. The Python checks exercise the link geometry, matrix identities, gauge covariance, curvature bounds, incidence constants, and frustrated negative controls. Numerical tests are not proofs of the theorems. Any Lean certificate is restricted to the stated scalar arithmetic; no analytic, gauge-theory, or continuum formalization is claimed. The first research pass and published repository sources are preserved. Literature context and precise source-access limits are in `audits/literature.md`.

References

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